

RAMYA AKULA, PH.D.

AI Research Scientist | Trustworthy AI · NLP · Computational Biology

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PROFESSIONAL SUMMARY

Ph.D.-trained AI researcher with 10+ years of experience spanning interpretable machine learning, natural language processing, computational biology, academic research, teaching, and international engineering. Developed peer-reviewed methods in interpretable AI and semantic evaluation, including work associated with U.S. Patent 12,210,833 and DARPA Forward Riser recognition. Recent postdoctoral research applies reproducible single-cell transcriptomic analysis to lung aging and disease. Experienced in translating complex scientific questions into rigorous computational methods, interpretable results, and collaborative research outputs.

RESEARCH & TECHNICAL STRENGTHS

Interpretable & Trustworthy AI Interpretability · Explainable ML · Model evaluation · Human-centered AI	NLP & Language Evaluation Transformers · Summarization · Semantic similarity · Figurative-language analysis	Computational Biology scRNA-seq · Scanpy · Seurat · Harmony · Differential expression
Machine Learning Deep learning · Representation learning · Transfer learning · Statistical evaluation	Scientific Computing Python · R · PyTorch · TensorFlow · Jupyter · Git · Linux · Docker	Research Translation Scientific communication · Mentoring · Customer discovery · Interdisciplinary collaboration

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar - Computational Biology & Machine Learning The Ohio State University, Department of Internal Medicine (Pulmonary) Columbus, OH, USA - Developed reproducible pipelines to integrate multi-cohort single-cell RNA-sequencing datasets, including quality control, Harmony-based batch correction, cell-type annotation, differential expression, and pathway-level interpretation. - Analyzed transcriptomic signatures of lung aging, cellular senescence, apoptosis, donor type, and chronic lung injury across human and mouse cohorts. - Partnered with pulmonary scientists to translate biological questions into analytical plans, interpretable visualizations, and publication-ready findings; contributed to a 2026 peer-reviewed publication in The Journals of Gerontology.	2024-2025
Lecturer - Computer Science & M.S. FinTech University of Central Florida Orlando, FL, USA - Designed and taught graduate and undergraduate courses in artificial intelligence, NLP, machine learning, data science, algorithms, blockchain, smart contracts, and programming. - Mentored and advised approximately 15 undergraduate students and supervised four senior projects in computer science.	2023-2024
Graduate Research Assistant - Trustworthy AI & NLP University of Central Florida, Complex Adaptive Systems Laboratory Orlando, FL, USA - Developed an interpretable multi-head self-attention architecture for sarcasm detection; the work received a U.S. patent and DARPA Forward Riser recognition. - Proposed SPEED Score, a sentence-embedding-based metric for evaluating abstractive and extractive summarization. - Researched explainable AI, figurative-language understanding, toxic-language detection, semantic evaluation, and AI for social good under DARPA-supported programs. - Served as NSF I-Corps Entrepreneurial Lead, conducting customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.	2018-2022
Project Engineer P3 Group GmbH Hamburg, Germany - Developed OSCAR (Offer-ability Specification Card Reader) for Airbus aircraft models. - Worked on Curved Stiffened Panel Sizing (CSPS) data extraction, including analysis of aircraft-part stress calculations, data extraction and visualization, and automated modification-closure reports for A350 ESI configuration management. - Developed a Java SE tool to convert ASCII data into user-readable PDF reports.	2015-2017
Master Thesis - Intelligent Visualization of Complex Social Networks Technical University of Kaiserslautern Kaiserslautern, Germany Implemented a visualization tool using hierarchical edge bundling and K-nearest neighbors to classify similar users, reduce visual clutter, and improve bundling strength in complex social networks.	2014-2015
Software Quality Assurance Analyst EXCO GmbH Mannheim, Germany Worked on ACCU-CHEK software for reading blood-glucose measurements from a medical device; developed and assessed software test cases.	2013-2014

Graduate Research Assistant	2013-2014
Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) GmbH Kaiserslautern, Germany	
Worked on 3D reconstruction for smart-society research: processed distorted images, calculated intrinsic and extrinsic camera parameters, and used camera calibration to estimate object rotation and displacement in the real world.	
Software Engineer	2011-2013
Regional Center for Urban and Environmental Studies (RCUES) Hyderabad, India	
- Developed widgets and responsive web applications for visualization of urban-planning statistics across desktop and mobile environments. - Developed a user-management system and a unified access point for datasets gathered from governmental organizations and ministries to improve administrative data retrieval.	
Software Developer	2010-2011
Inventurus Software Solutions Hyderabad, India	
Developed a Java embedded-database application for health-insurance data management and optimized dynamic search across text and scanned documents using pattern matching and feature extraction.	
Bachelor Thesis - Chat Client	2009-2010
Satya Technologies Hyderabad, India	
Implemented a Java Applet chat client with multithreaded message processing and a chat-export feature for maintaining local conversation history.	

SELECTED RESEARCH, INNOVATION & RECOGNITION

- U.S. Patent 12,210,833 (2025): Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection.
- DARPA Forward Riser (2022): Recognition associated with research in interpretable machine learning for figurative-language understanding.
- NSF I-Corps Entrepreneurial Lead (2019): Customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.
- First Place, Startup Weekend Orlando (2021): Contributed to an AI-powered online youth-safety product concept.
- Research coverage listed in the prior CV includes NBC News, IEEE Spectrum, Los Angeles Times, Defense One, ScienceDaily, and ABC Australia.

SELECTED PUBLICATIONS

- Akula, R., et al. "Transcriptomic shifts in human lungs donated after brain and circulatory death and severe Obliterative Bronchiolitis." The Journals of Gerontology, 2026.
 - Akula, R., & Garibay, I. "SPEED Score: Sentence Pair Embeddings Based Evaluation Metric for Abstractive and Extractive Summarization." LREC, 2022.
 - Akula, R., & Garibay, I. "Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media." Entropy, 2021.
 - Akula, R., & Garibay, I. "Explainable Detection of Sarcasm in Social Media." ACL Workshop, 2021.
- [Complete publication record on Google Scholar](#)

EDUCATION

Ph.D. in Computer Science	2018-2022
University of Central Florida Orlando, FL, USA Advisor: Dr. Ivan Garibay	
M.S. in Computer Science	2013-2015
Technical University of Kaiserslautern Germany Advisor: Dr. Hans Hagen	
B.Tech. in Computer Science and Engineering	2006-2010
Jawaharlal Nehru Technological University India	